

Current Mirror – Layout Design Report

ADI Internship

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1. Objective

This report documents the physical (layout) design of a transistor NMOS current mirror used to bias two branches (I1, I2) from a common diode-connected reference. It covers the device placement strategy, metal-layer routing (M1–M3), and layout verification (DRC / LVS).

2. Schematic and Device Sizing

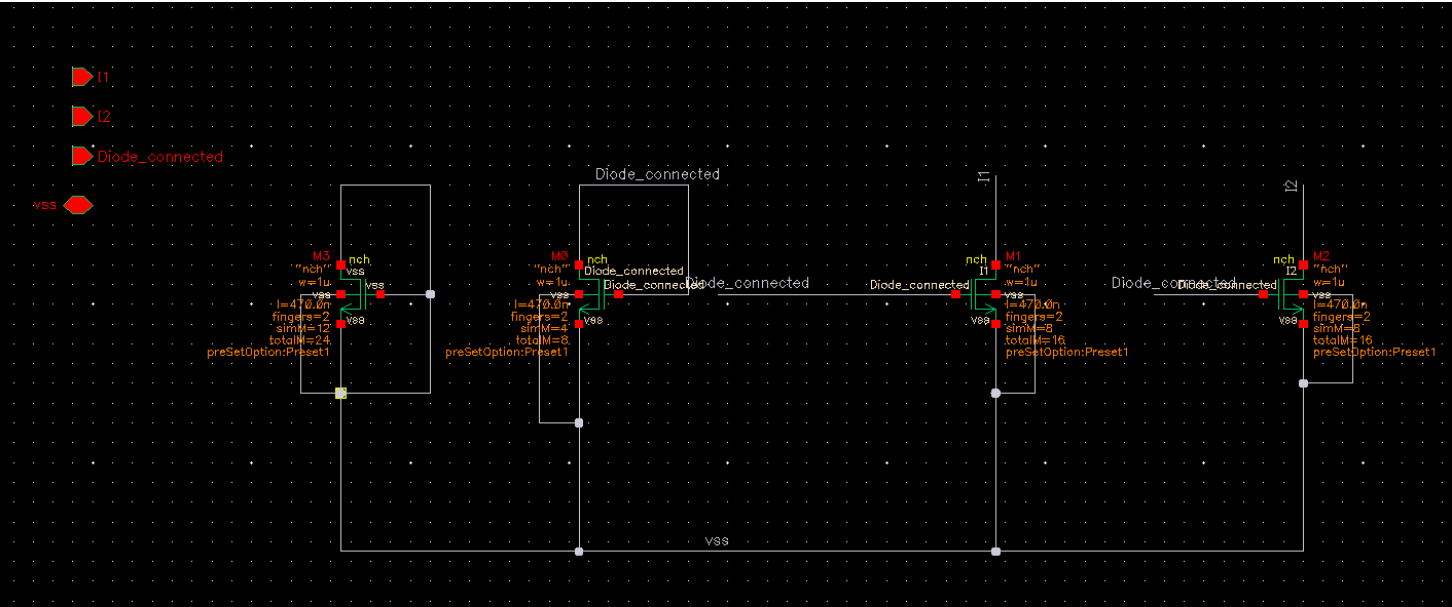


Figure 1. Current mirror schematic — reference devices M0 (diode-connected) and output branches M1 (I2), M2 (I1), all NMOS, common source at VSS.

All four devices share W = 1 μ m, L = 470 nm, 2 fingers; multiplier is scaled per branch (M0 $\times$ 4, M1 $\times$ 8, M2 $\times$ 8, M3 $\times$ 12, i.e. total multiplicity 4/8/8/12) to set the 1:2:2 current mirroring ratio between I1 and I2 and M3 is dummy.

Table 1. Device sizing summary read from the schematic.

Device	Type	WF	L	Fingers	Mult. (sim)
M0	nch (diode-conn.)	1 μ m	470 nm	2	4
M1	nch (mirror, I1)	1 μ m	470 nm	2	8
M2	nch (mirror, I2)	1 μ m	470 nm	2	8
M3	nch (dummy)	1 μ m	470 nm	2	12

3. Layout Floorplan and Matching Pattern

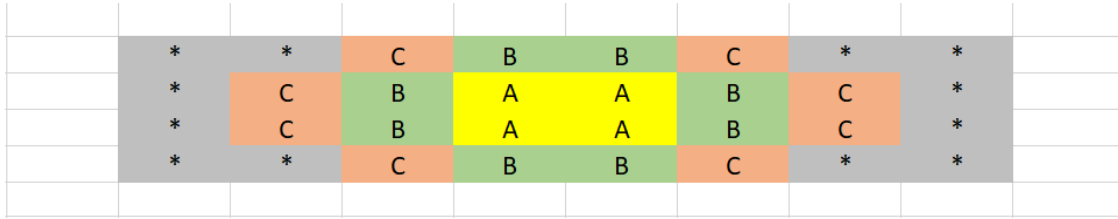


Figure 2. Common-centroid pattern used for the mirror devices (A/B/C unit-cell arrangement with dummy '*' columns on both edges).

Centroid symmetry minimizes gradient-induced mismatch between the reference and output branches; dummy columns keep etch/diffusion boundary conditions identical for all active fingers.

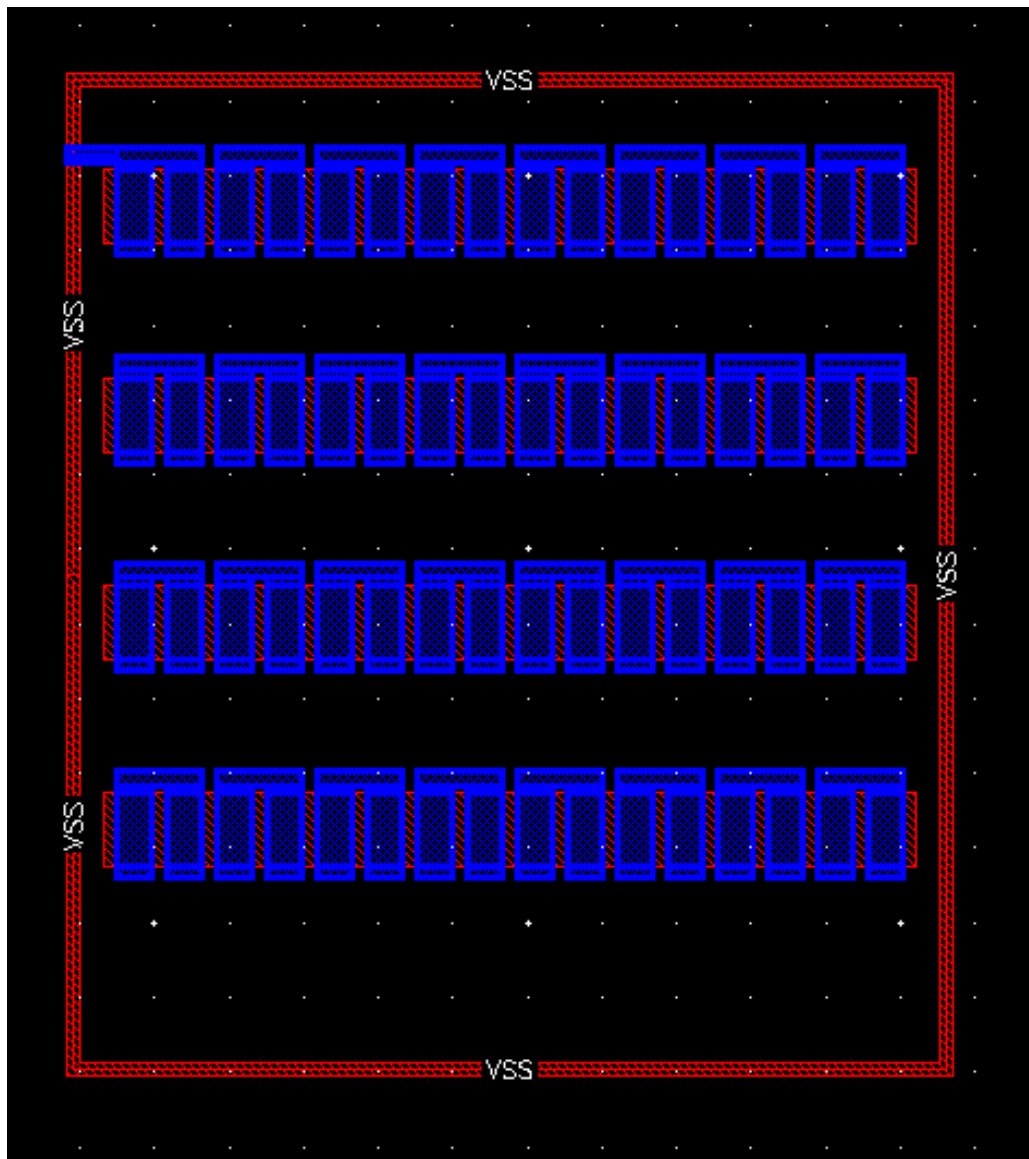


Figure 3. Device placement — M0–M3 unit transistors arranged per the pattern before metal routing.

Placement follows Fig. 2 directly; no routing has been added yet at this stage.

4. Metal Routing

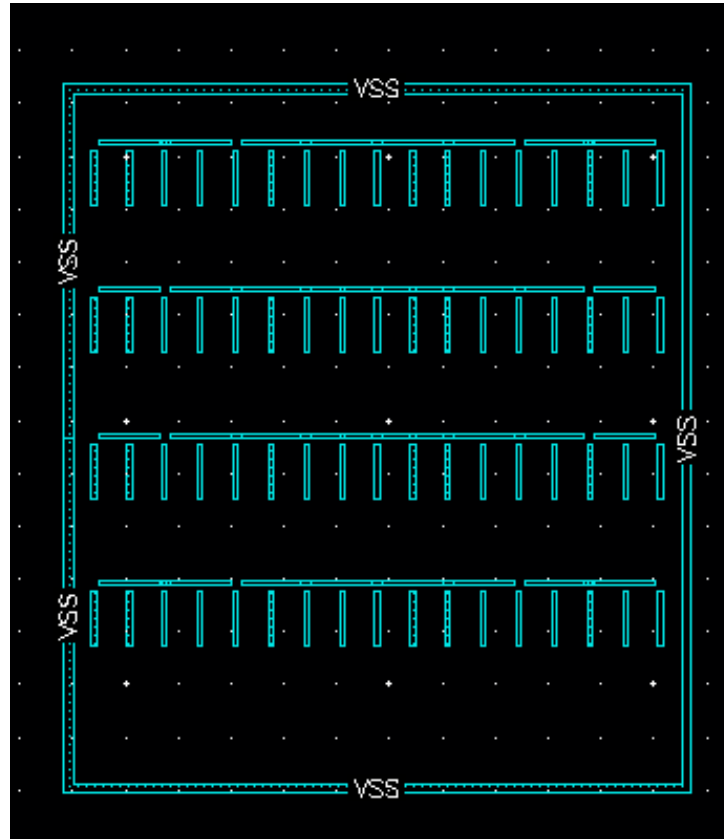


Figure 4. M1 (metal 1) routing layer — local source/drain/gate interconnect for the unit transistors.

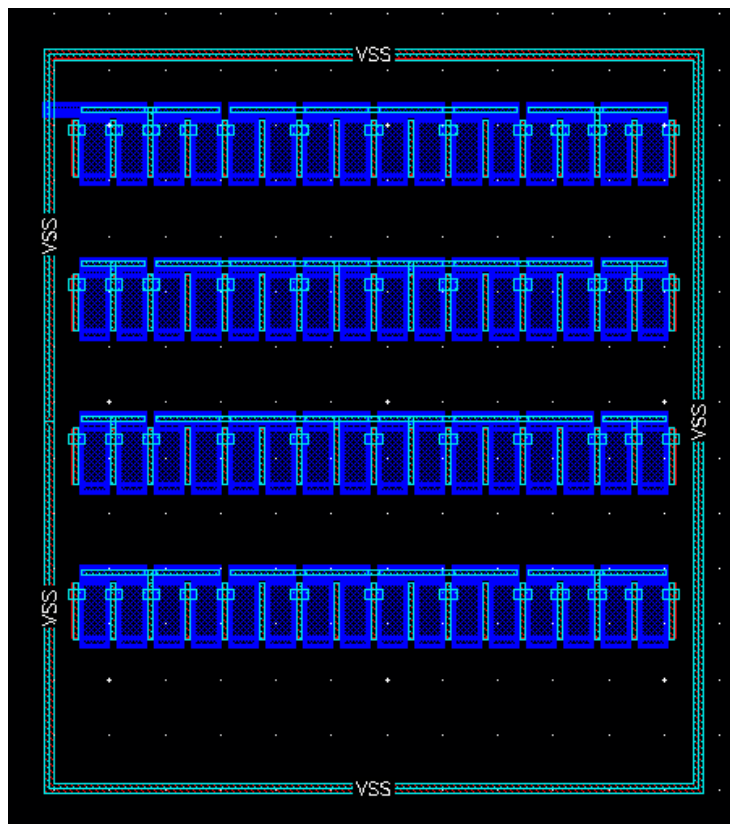


Figure 5. Layout after adding the M1 layer.

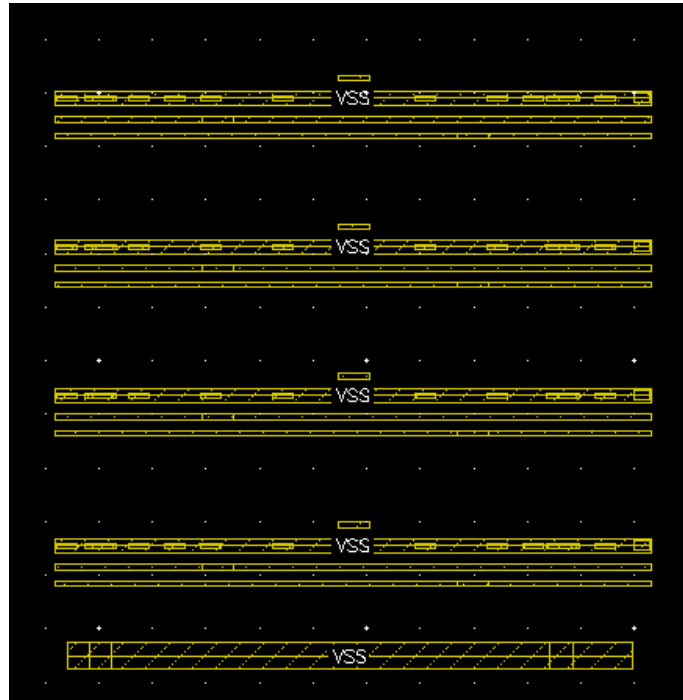


Figure 6. M2 (metal 2) routing layer — branch-level interconnect tying unit transistors of each device together.

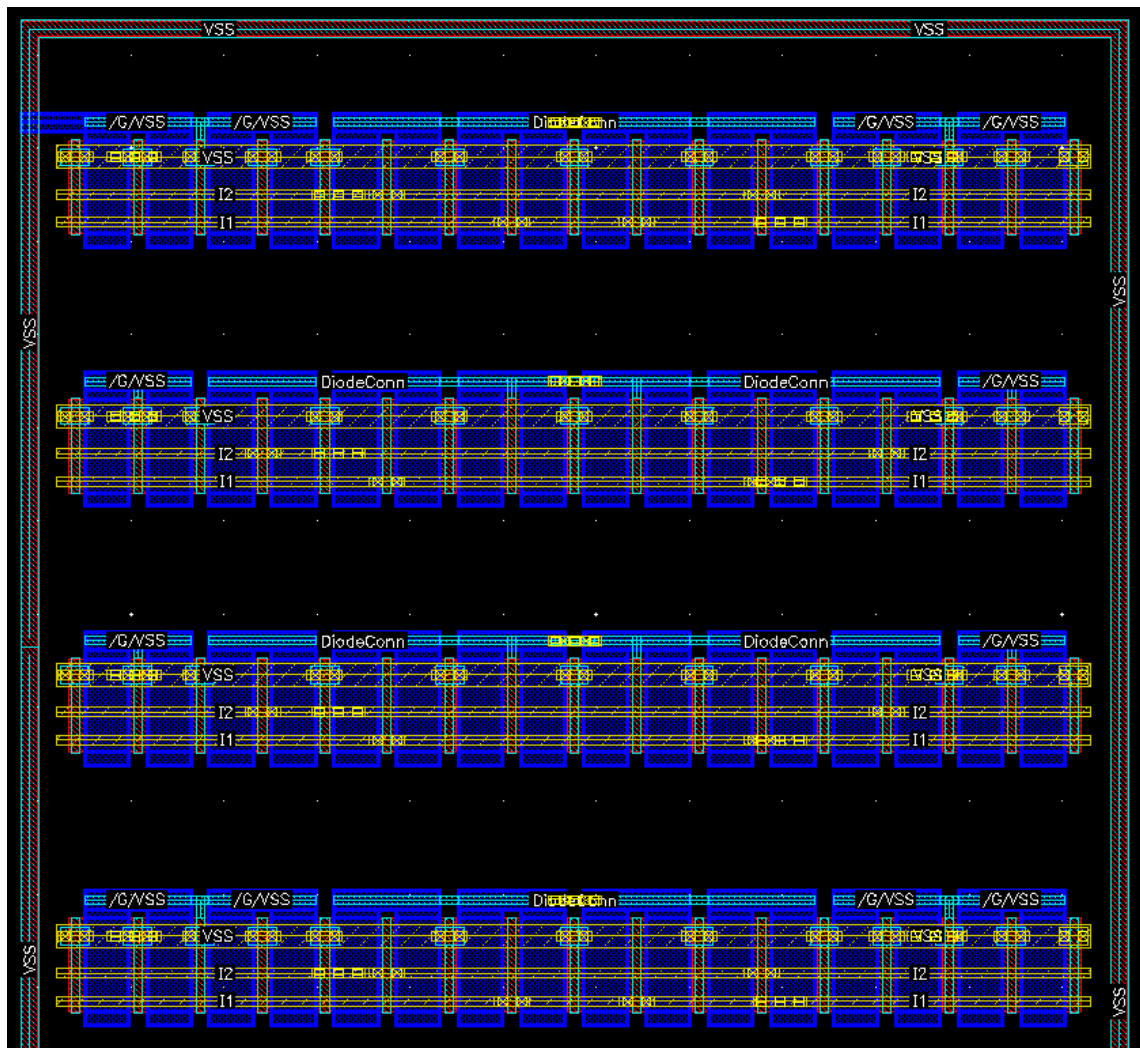


Figure 7. Layout after adding the M2 layer.

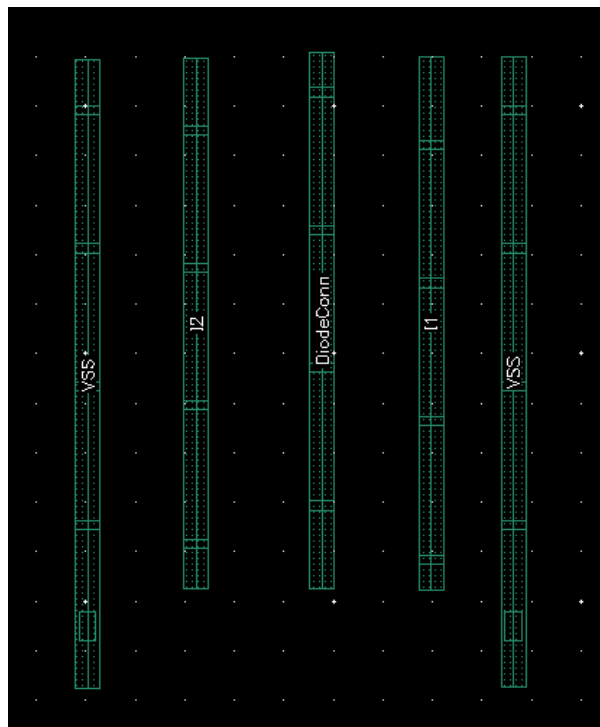


Figure 8. M3 (metal 3) routing layer — top-level routing to the I1, I2 outputs and VSS.

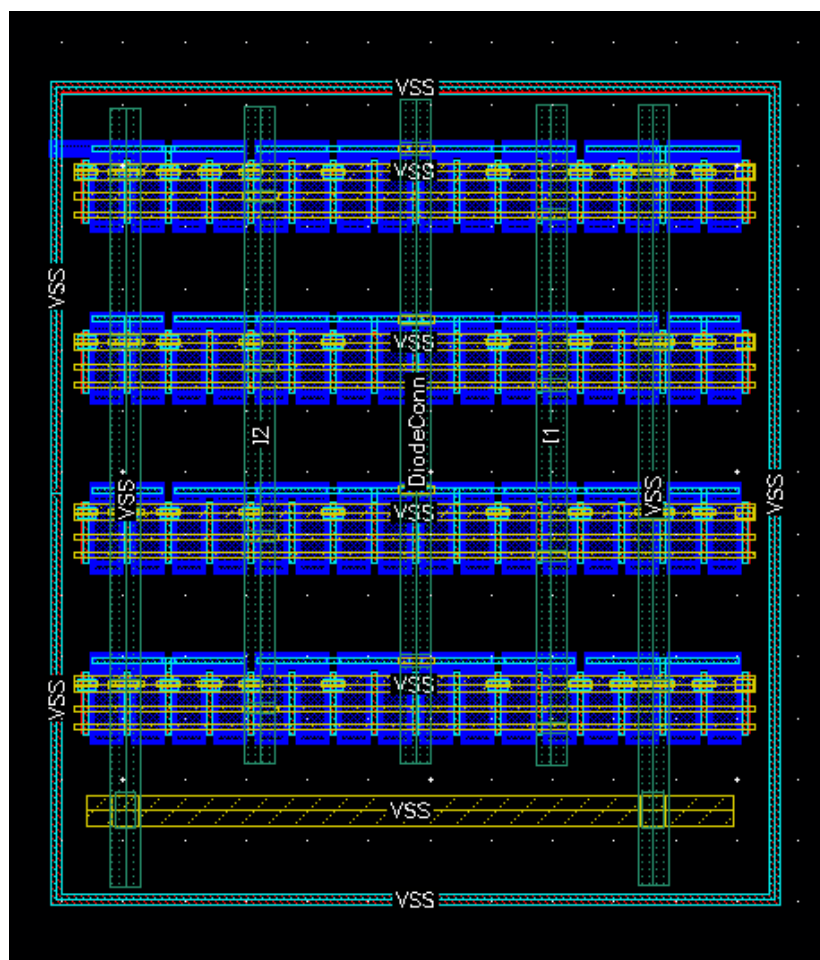


Figure 9. Layout after adding the M3 layer.

Routing complete; block is ready for boundary closure and verification.

5. Layout Verification



Figure 10. DRC (Design Rule Check) result.

DRC test passed — layout is clean against the foundry design-rule deck (no spacing/width/enclosure violations).

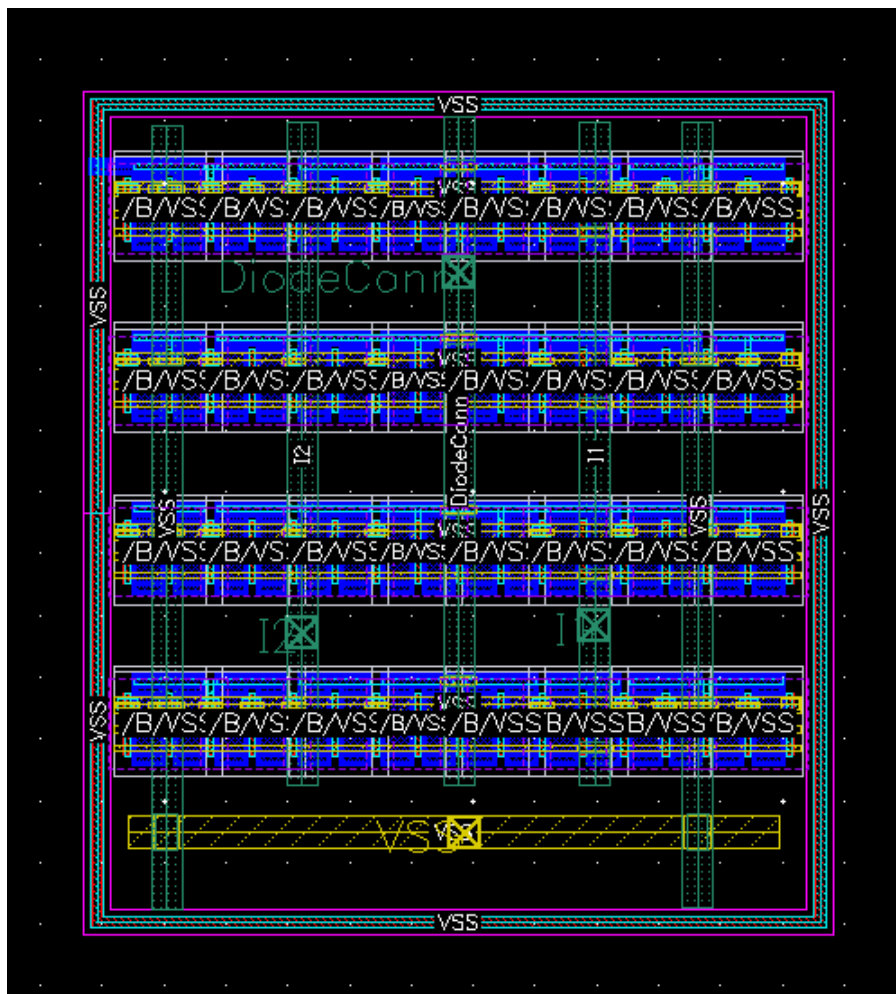


Figure 11. Final layout of the current mirror after boundary closure and pin labeling. Compact, symmetric layout consistent with the common-centroid floorplan of Fig. 2.

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#####
#          CALIBRE SYSTEM          #
#          LVS REPORT              #
#####

REPORT FILE NAME:  CM lvs report
LAYOUT NAME:      /home/cadence/Desktop/PDK_CRN6SLP_v1_7a_Official_IC61_20120914_all/PDK_CRN6SLP_v1_7a_Official_IC61_20120914/Calibre_Cache_Trash/CM.sp ('CM')
SOURCE NAME:      /home/cadence/Desktop/PDK_CRN6SLP_v1_7a_Official_IC61_20120914_all/PDK_CRN6SLP_v1_7a_Official_IC61_20120914/Calibre_Cache_Trash/CM.src.net ('CM')
FILE FILE:        /home/cadence/Desktop/PDK_CRN6SLP_v1_7a_Official_IC61_20120914_all/PDK_CRN6SLP_v1_7a_Official_IC61_20120914/Calibre_Cache_Trash/_calibre.lvs_
CREATION TIME:    Wed Jul 22 03:51:22 2026
CURRENT DIRECTORY: /home/cadence/Desktop/PDK_CRN6SLP_v1_7a_Official_IC61_20120914_all/PDK_CRN6SLP_v1_7a_Official_IC61_20120914/Calibre_Cache_Trash
USER NAME:        cadence
CALIBRE VERSION:  v2014.1_17.12   Thu Feb 27 15:53:54 PST 2014

OVERALL COMPARISON RESULTS

#          #####          T  T
#          # CORRECT      #  |
#          #              #  /
#          #####

=====
CELL SUMMARY
=====
Result      Layout      Source
-----
CORRECT     CM             CM

=====
LVS PARAMETERS
=====

o LVS Setup:
// LVS COMPONENT TYPE PROPERTY
// LVS PINNUMBER CHANGE SENSITIVITY

```

Figure 12. LVS (Layout vs. Schematic) result.

LVS test passed — extracted layout netlist matches the schematic of Fig. 3 (devices, connectivity, and node count).

6. Conclusion

- Interdigitated, common-centroid placement was used to minimize mirror mismatch between I1 and I2.
- Routing was built up and checked layer-by-layer (M1 → M2 → M3) to isolate errors early.
- Final layout passed both DRC and LVS, confirming the physical implementation matches the intended schematic.